Reproducing the Walert retrieval evaluation

Results summary — Australian Casual Employee Work-Rights RAG Assistant**. Source notebook:** `Walert_Reproduction_Analysis.ipynb` **in this repository. Every figure below is produced by a cell in that notebook; run it top to bottom to regenerate them.**

Setup

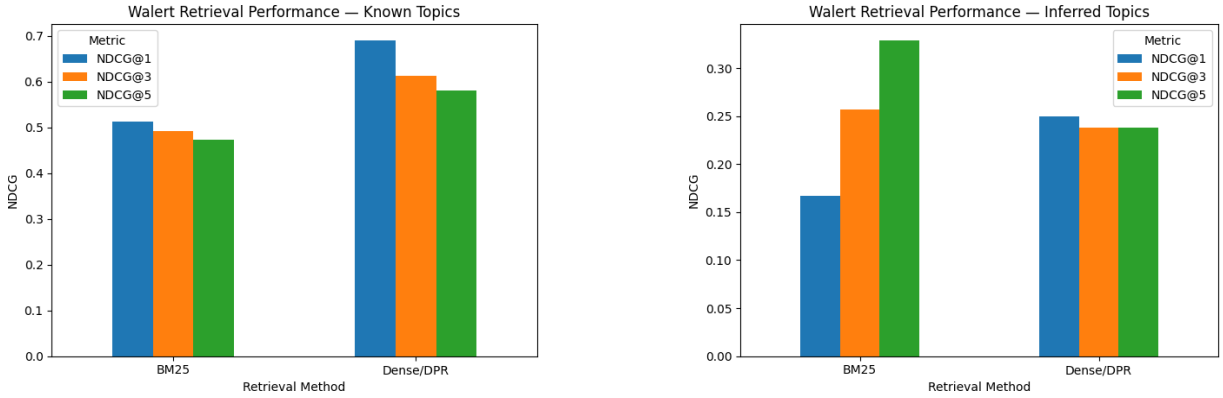
Python 3.9.6, NumPy 1.24.3, pandas 2.0.3, ranx. Collection loads as 106 questions, 120 passages, 140 ground-truth rows and 500 relevance judgments; released runs contain 7,099 lines (BM25) and 10,600 (dense). Scoring uses the authors' `src/retrieval/eval.py`.

1. Released runs re-scored against the paper

Questions	System	@1	paper	diff	@3	paper	diff	@5	paper	diff
Known	BM25	0.5119	0.512	−0.0001	0.4912	0.491	+0.0002	0.4733	0.473	+0.0003
Known	Dense	0.6905	0.691	−0.0005	0.6119	0.612	−0.0001	0.5812	0.581	+0.0002
Inferred	BM25	0.1667	0.167	−0.0003	0.2566	0.256	+0.0006	0.3291	0.329	+0.0001
Inferred	**Dense**	**0.2500**	**0.250**	**0.0000**	**0.2380**	**0.235**	**+0.0030**	**0.2380**	**0.235**	**+0.0030**

Ten of twelve cells agree to within 0.001, the finest resolution the paper's three-decimal reporting allows. The two that do not are one disagreement counted twice: the dense run returns the same ranking at cutoffs 3 and 5, so 0.2380 against a reported 0.235 appears at both. Every other cell agrees to within 0.0006. The inferred set is small, so one question changing rank shifts the mean by about this much; a tie-breaking difference in the installed ranx is the likely cause, untested.

2. Reproduced effectiveness by question type



Figures as produced by the notebook. Each panel is scaled to its own data range (known reaches 0.7, inferred 0.33), so compare by value rather than bar height. Dense leads BM25 at every cutoff on known topics; on inferred topics both fall by more than half and the ordering reverses with depth, BM25 overtaking at cutoffs 3 and 5.

3. Independent BM25 reimplementation

Repository history shows three commits adding run files, so the runs are committed artefacts and re-scoring them tests the evaluation harness rather than the retrieval pipeline. BM25 was reimplemented over the 120-passage collection with `rank_bm25`, tokenising by lowercasing and whitespace splitting, writing 1,060 rows and scoring against the same judgments.

Questions	Run	NDCG@1	NDCG@3	NDCG@5
Known	my.bm25 (independent)	0.5000	0.4389	0.3919
Known	walert.rag.bm25 (released)	0.5119	0.4912	0.4733
Inferred	my.bm25 (independent)	0.1667	0.1098	0.1588
Inferred	walert.rag.bm25 (released)	0.1667	0.2566	0.3291

The two runs agree closely at rank 1 — 0.5000 against 0.5119 on known, identical at 0.1667 on inferred — and diverge with depth, most sharply on inferred questions. Agreement at rank 1 indicates the ranking function and collection are handled correctly; the divergence points to text preprocessing, since this implementation applies no stemming or stopword removal while standard BM25 toolkits apply both. A tokenisation ablation to confirm this is the next step.

Findings

- The published retrieval evaluation reproduces, with one unexplained 0.0030 discrepancy on the dense run.
- Effectiveness drops by more than half from known to inferred questions under every configuration measured, including the independent implementation. That gap, not peak accuracy on well-formed queries, is the binding constraint for a system answering questions phrased in everyday language.
- Re-scoring committed run files verifies scoring only. Independent reimplementation is what tests the pipeline.

Scope and limitations

Generation metrics (BERTScore, ROUGE-1, and the share of out-of-knowledge-base questions correctly withheld) are not covered and require running an LLM over retrieved passages. The intent-based baseline is committed to the repository but was not scored, so the paper's central intent-versus-RAG trade-off is not assessed here and the significance comparison family differs from the published one. Only BM25 was reimplemented; the dense run remains as committed. The preprocessing and tie-breaking explanations above are hypotheses consistent with the results, not tested findings.